12.2 Graphs and surfaces ; 12.3 Contour diagrams

The surface in $\mathbb{R}^3$ defined by an equation is the set of solutions (x,y,z) to that equation.

The graph of a function f of two variables is the set of points (x,y,z) in $\mathbb{R}^3$ such that $z = f(x,y)$.

The cross section of f with $x=c$ is the intersection of the graph with the plane $x=c$. We can also intersect with planes $y=c$ and $z=c$.

→ Geogebra demo, do problems 1,2

Tip: If a surface in $\mathbb{R}^3$ is defined by an equation with a missing variable, e.g., $y^2+z^2=1$, then the cross sections with the missing variable fixed are all the same.

→ Do problem 3

Contour lines of a surface are the cross sections with z fixed. A contour diagram is a collection of level curves with labeled z values.

→ Do problems 4-7.